Daifeng Li

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EDUCATION

The Hong Kong University of Science and Technology (HKUST) Sep. 2026-Present
Ph.D. student in Computer Science and Engineering, Hong Kong SAR.
Advisor: Prof. Binhang Yuan.

University of Science and Technology of China (USTC) Sep. 2022–Jul. 2026
B.S. in Computer Science, Huaxia Computer Talent Program, Hefei, China.
GPA: 3.9/4.3; Weighted Average: 90.74/100; Rank: Top 5% (10/213).

Coursework Operating Systems (Honors) (99/100, Rank 1st); Computer Organization and Design (Honors) (98/100); Fundamentals of Algorithmics (96/100); Analog and Digital Circuits (94/100).

RESEARCH INTERESTS

Areas Machine learning systems; GPU/CUDA programming; LLM-based code generation; compiler and programming systems; reinforcement learning systems.

RESEARCH AND INDUSTRY EXPERIENCE

Research Intern, Alibaba Qwen Team Mar. 2026–Present

LLM-Generated CUDA Kernel Optimization

- First author of **D2K-Bench**, released during my internship with the Alibaba Qwen Team.
- Research LLM-based CUDA kernel generation and GPU operator optimization, connecting model training, compiler feedback, and executable kernel performance.
- Participated in Qwen3.7-Max training and evaluation; CUDA-kernel writing was selected as an important showcased model capability.
- Evaluated Qwen3.7-Max on KernelBench, reporting speedups of 2.27× geometric mean and 2.30× median, with generated kernels faster than eager and compiled baselines in most cases.

ARGUS: Agentic GPU Optimization Guided by Data-Flow Invariants 2025–2026

Core Contributor; Kernel Generation and Compiler-Coupled Optimization

- Core contributor to ARGUS, an agentic GPU optimization framework guided by data-flow invariants for LLM-based kernel generation.
- Implemented and optimized FlashAttention kernels on AMD GPUs, focusing on program construction, memory access behavior, and performance-oriented kernel design.
- Contributed MoE and attention operators to vLLM/SGLang on AMD GPUs, improving support for high-performance inference workloads.
- Completed major benchmark execution, ablation analysis, and performance comparison across generated GPU kernels.
- Helped design the kernel-generation and compiler-interaction pipeline for reliable, high-performance GPU optimization.

SELECTED PUBLICATIONS

1. **Daifeng Li**, Huiqiang Jiang, Chengruidong Zhang, Wei Wu, Xudong Guo, Jianhong Tu, Jianwei Zhang, Binhang Yuan, and Dayiheng Liu. *D2K-Bench: Can LLM Agents Turn Expert Designs into Efficient GPU Kernels?*. arXiv preprint arXiv:2610.03226, 2026.
2. Haohui Mai, Xiaoyan Guo, Xiangyun Ding, **Daifeng Li**, Qiuchu Yu, Chenzhun Guo, Cong Wang, Jiacheng Zhao, Christos Kozyrakis, and Binhang Yuan. *Ave: Guiding Agentic GPU Optimization Using Data-Flow Invariants*. ACM Symposium on Operating Systems Principles (SOSP), 2026.
3. Jiarui Zhang, Yuchen Yang, Ran Yan, Zhiyu Mei, Liyuan Zhang, **Daifeng Li**, Wei Fu, Jiaxuan Gao, Shusheng Xu, Yi Wu, and Binhang Yuan. *AReaL-DTA: Dynamic Tree Attention for Efficient Reinforcement Learning of Large Language Models*. International Conference on Machine Learning (ICML), 2026.

PROJECT EXPERIENCE

National Student Computer System Capability Challenge Fall 2024
National Third Prize (Team Award), China

- Designed an advanced out-of-order, four-issue CPU for the national computer system capability challenge.
- Implemented key microarchitectural components including cache, bus systems, out-of-order queues, computational units, and multiplication/division units.
- Applied computer architecture techniques including multi-issue execution and memory hierarchy optimization.

SKILLS

Programming C, C++, Python, CUDA, Triton, Verilog, Chisel, x86 Assembly.
ML / Systems PyTorch, TensorFlow, GPU kernel optimization, LLM-based code generation, compiler-guided optimization.
Tools Git, GDB, LaTeX, Linux, performance benchmarking and profiling.

TEACHING EXPERIENCE

Teaching Assistant, Computer Programming USTC, 2024
Teaching Assistant, Operating Systems (Honors) USTC, 2025
Teaching Assistant, Computer Programming USTC, 2025

HONORS AND AWARDS

- National Scholarship, Ministry of Education of China, 2024 and 2025. Awarded twice; among the most selective undergraduate scholarships in China, recognizing top academic performance and overall distinction.
- Soong Ching Ling Future Scholarship, 2023 and 2024. Awarded twice for academic excellence and undergraduate research potential.
- National Student Computer System Capability Challenge, National Third Prize (Team Award), China, 2024. Built an advanced out-of-order, four-issue CPU in a national computer systems competition.
- Endeavor Scholarship, University of Science and Technology of China, 2023. University-level scholarship recognizing strong academic performance in the Huaxia Computer Talent Program.